

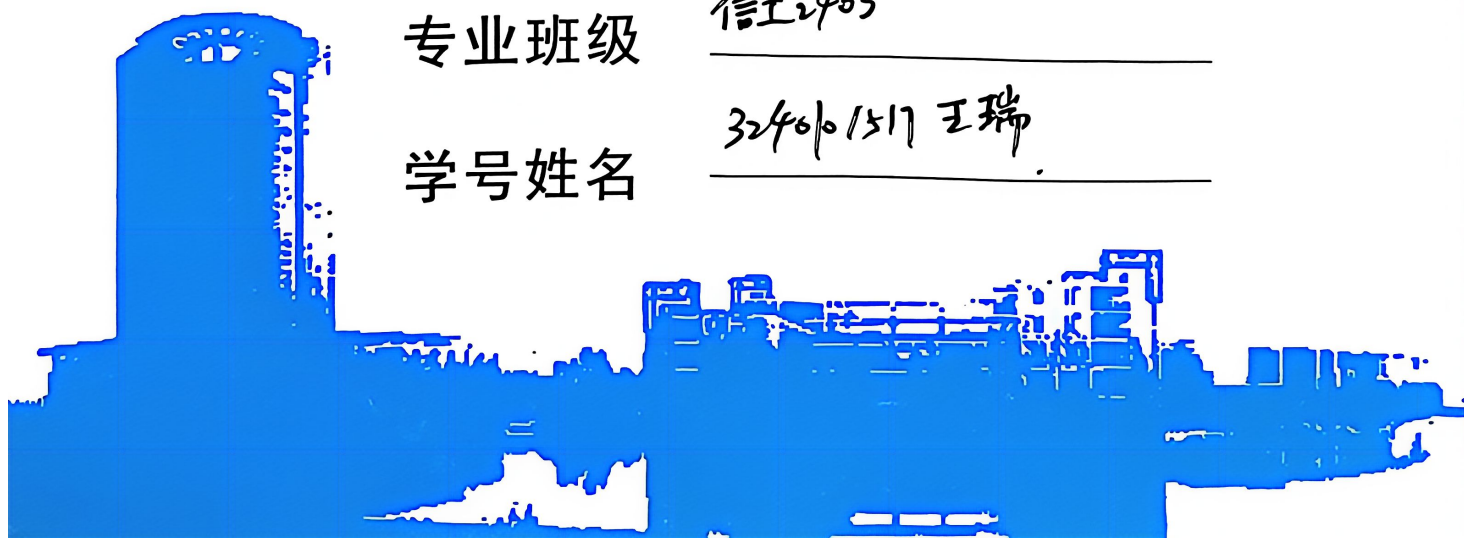
浙江大学

Exercise Book

课程名称 数字系统设计

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春周作业

1. 非门: $\bar{x} = x \oplus 1$

或门已有

与门 $xy = \overline{\bar{x} + \bar{y}}$

因此异或门、或门可以实现与门或门与非门

因而是完备的

2. $Y = (A\bar{B}B + A\bar{B} + ABD + D\bar{B})D + AE(B + \bar{D})$

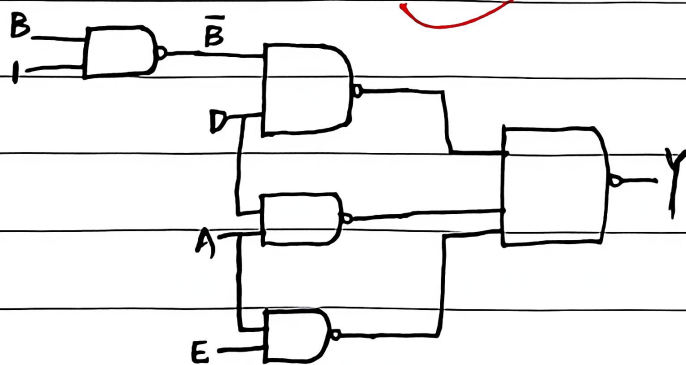
$$= A\bar{B}D + A\bar{B}D + \bar{B}D + AE(B + \bar{D})$$

$$= AD + \bar{B}D + AEB + AED$$

$$= A(D + \bar{D}E) + \bar{B}D + AEB$$

$$= AD + AE + \bar{B}D$$

$$= \overline{\bar{A}D \cdot \bar{A}E \cdot \bar{B}D}$$



3. $b_0, b_1, b_2, b_3 \in \{0, 1\}$ 十进制数 $X = 6b_3 + 3b_2 + b_1 + b_0$

十进制 二进制码

0 0000

1 0001

2 0011

3 0100

4 0101

5 0111

6 1000

7 1001

8 1011

9 1100

4.

二进制

循环码

二进制

循环码

000

000

101

111

001

001

110

101

010

011

110

100

011

010

111

100

100

110

5. 1100101 110110

BCD 01010100 $\rightarrow$ 54 (十进制) $\rightarrow$ 110110 二进制

6. 0100

7. $A_0 \ A_1 \ A_2 \ A_3 \ Y$

0 0 0 0 0

$$Y = A_0 \bar{A}_1 \bar{A}_2 \bar{A}_3 + \bar{A}_0 A_1 \bar{A}_2 \bar{A}_3 + \bar{A}_0 \bar{A}_1 A_2 \bar{A}_3$$

1 0 0 0 1

$$+ A_0 A_1 A_2 \bar{A}_3 + \bar{A}_0 \bar{A}_1 \bar{A}_2 A_3$$

0 1 0 0 1

$$+ A_0 A_1 \bar{A}_2 A_3 + A_0 \bar{A}_1 A_2 A_3$$

1 1 0 0 0

$$+ \bar{A}_0 A_1 A_2 A_3$$

0 0 1 0 1

1 0 1 0 0

0 1 1 0 0

1 1 1 0 1

0 0 0 1 1

1 0 0 1 0

0 1 0 1 0

1 1 0 1 1

0 0 1 1 0

1 0 1 1 1

0 1 1 1 1

1 1 1 1 0

8.

(1)

AB \ CD	00	01	11	10
00	1	1		X
01	1	X	1	
11	X	1	X	X
10	X	1		1

$$F(A, B, C, D) =$$

AB \ CD	00	01	11	10
00	1	1		X
01	1	X	1	
11	X	1	X	X
10	X	1		1

$$F(A, B, C, D) = \bar{C} + BCD + A\bar{C}\bar{D}$$

$$2) Y(A, B, C, D) = \Sigma m(1, 3, 4, 6, 7, 9, 11, 12, 14, 15)$$

$$= \Sigma m(0, 2, 5, 8, 10, 13) \neq$$

AB \ CD	00	01	11	10
00	1			1
01		1		
11			1	
10	1			1

$$Y(A, B, C, D) = \bar{B}\bar{D} + B\bar{C}D$$

2)

AB \ CD	00	01	11	10
00	1	1		0
01	0	0	0	0
11	0	0	1	1
10	1	1	1	1

$$Y(A, B, C, D) = \bar{B}\bar{C}\bar{D} + A\bar{C}D + BCD + A\bar{B}C$$

$$\bar{Y} = A\bar{B}\bar{C}\bar{D} + \bar{A}B + \bar{A}\bar{B}\bar{D} + B\bar{C} + BCD$$

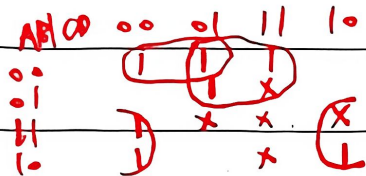
$$Y = \bar{A}\bar{B}D + A\bar{C}\bar{D} + A\bar{B}D = \bar{B}D + A\bar{C}\bar{D}$$

9.

$$1) Y = \bar{A}D + AB\bar{C} + A\bar{B}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D}$$

$$ABC, ABD, ACD, BCD = 0$$

$$= \bar{A}D + AB + A\bar{B}\bar{D} + \bar{A}\bar{B}\bar{C}\bar{D}$$



$$Y = A\bar{D} + \bar{A}D + A\bar{B}\bar{C}$$

$$2) Y = \sum m(0, 2, 5, 8, 10, 13)$$

$$= \bar{B}\bar{D} + B\bar{C}D$$

$$10 \quad \bar{B}D \quad x0x1 \quad 000|_6=1 \quad 001|_6=3 \quad 010|_6=9 \quad 101|_6=11$$

$$\bar{A}D \quad 0xx1 \quad 010|_6=5 \quad 011|_6=7$$

$$BD \quad x1x1 \quad 110|_6=13 \quad 111|_6=15$$

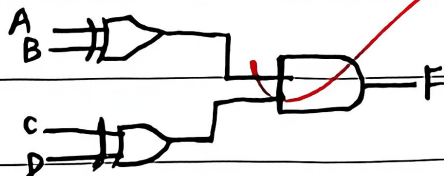
$$F = \sum m(1, 3, 5, 7, 9, 11, 13, 15)$$

$$= \pi M(0, 2, 4, 6, 8, 10, 12, 14)$$

$$11. \quad F = (\bar{A}B\bar{C} + A\bar{B}\bar{C})\bar{D} + (A\bar{B}\bar{C} + \bar{A}B\bar{C})D$$

$$= (A \oplus B)C\bar{D} + (A \oplus B)\bar{C}D$$

$$= (A \oplus B)(C \oplus D)$$



12.

$$F_1(A, B, C, D) = \bar{C}D + \bar{A}CD + ABC\bar{D}$$

$$F_2(A, B, C, D) = CD + BD$$

~~$$F_3(A, B, C, D) = F_1 + F_2 + F$$~~

A	B	C	D	F ₁	F ₂	F ₃	F ₃
0	0	0	0	0	0	0	00
0	0	0	1	1	0	1	01
0	0	1	0	0	0	0	11
0	0	1	1	1	1	0	10
0	1	0	0	0	0	0	00
0	1	0	1	1	1	0	01
0	1	1	0	0	0	0	11
0	1	1	1	1	1	0	10
1	0	0	0	0	0	0	00
1	0	0	1	1	0	1	01
1	0	1	0	0	0	0	11
1	0	1	1	0	0	0	10
1	1	0	0	0	0	0	00
1	1	0	1	1	1	0	01
1	1	1	0	1	0	1	11
1	1	1	1	0	1	1	10

$$F_3(A, B, C, D)$$

$$= \bar{A}\bar{B}\bar{C}D + A\bar{B}D + ABC$$

$$= \bar{B}D(\bar{A}\bar{C} + A) + ABC$$

$$= \bar{B}D(\bar{C} + A) + ABC$$

$$= ABC + A\bar{B}D$$

$$+ \bar{B}\bar{C}D$$

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